



LeetCode 1572: Matrix Diagonal Sum

1. Problem Title & Link

Title: LeetCode 1572: Matrix Diagonal Sum

Link: <https://leetcode.com/problems/matrix-diagonal-sum/>

2. Problem Statement (Short Summary)

Given a square matrix:

mat

Return the sum of:

- Primary Diagonal
- Secondary Diagonal

Do **not count the same element twice** if both diagonals intersect.

Example

Input:

```
[
  [1,2,3],
  [4,5,6],
  [7,8,9]
]
```

Primary diagonal:

$1 + 5 + 9$

Secondary diagonal:

$3 + 5 + 7$

Total:

$1+5+9+3+5+7 = 30$

But:

5

belongs to both diagonals.

Count it once.

Answer:

25

3. Examples (Input → Output)

Example 1

Input:

```
[
```



```
[1,2,3],  
[4,5,6],  
[7,8,9]  
]
```

Output:

25

Example 2

Input:

```
[  
[1,1,1,1],  
[1,1,1,1],  
[1,1,1,1],  
[1,1,1,1]  
]
```

Output:

8

Example 3

Input:

```
[  
[5]  
]
```

Output:

5

4. Constraints

$n == \text{mat.length}$

$n == \text{mat}[i].\text{length}$

$1 \leq n \leq 100$

$1 \leq \text{mat}[i][j] \leq 100$

Important:

Matrix is always square.



5. Core Concept (Pattern / Topic)

Matrix Traversal

Secondary Topics:

- Index Pattern Recognition
- Diagonal Traversal

6. Thought Process (Step-by-Step Explanation)

Understanding Diagonals

Primary Diagonal

Positions:

(0,0)

(1,1)

(2,2)

(3,3)

Pattern:

row == column

Secondary Diagonal

Positions:

(0,n-1)

(1,n-2)

(2,n-3)

...

Pattern:

row + column = n - 1

Example

Matrix:

1 2 3

4 5 6

7 8 9



Primary:

1
5
9

Secondary:

3
5
7

Total

$1 + 5 + 9 + 3 + 5 + 7$

But:

5
counted twice.

Subtract once.

Answer:

25

Key Observation

For every row:

Primary diagonal element:

$\text{mat}[i][i]$

Secondary diagonal element:

$\text{mat}[i][n-1-i]$

Center Element Problem

Occurs only when:

n is odd

Example:

3 x 3

Center:

(1,1)

belongs to both diagonals.

Need to subtract once.

7. Visual / Intuition Diagram (ASCII Diagram)



Example:

```
1 2 3
4 5 6
7 8 9
```

Primary:

```
X . .
. X .
. . X
```

Secondary:

```
. . X
. X .
X . .
```

Combined:

```
X . X
. X .
X . X
```

Center:

```
5
appears twice.
Subtract once.
```

8. Pseudocode (Language Independent)

```
sum = 0

for i from 0 to n-1

    sum += mat[i][i]

    sum += mat[i][n-1-i]

if n is odd

    subtract center element

return sum
```

9. Code Implementation

Python

```
class Solution:
    def diagonalSum(self, mat):

        n = len(mat)

        total = 0
```



```
for i in range(n):  
    total += mat[i][i]  
    total += mat[i][n - 1 - i]  
  
if n % 2 == 1:  
    total -= mat[n // 2][n // 2]  
  
return total
```

Java

```
class Solution {  
    public int diagonalSum(  
        int[][] mat) {  
        int n = mat.length;  
        int total = 0;  
        for(int i = 0; i < n; i++) {  
            total += mat[i][i];  
            total += mat[i][n - 1 - i];  
        }  
        if(n % 2 == 1) {  
            total -= mat[n / 2][n / 2];  
        }  
        return total;  
    }  
}
```

10. Time & Space Complexity

Time Complexity

$O(n)$

Only one traversal through rows.

Space Complexity

$O(1)$



No extra data structures.

11. Common Mistakes / Edge Cases

Mistake 1

Double counting center element.

Example:

1 2 3

4 5 6

7 8 9

Answer becomes:

30

instead of:

25

Mistake 2

Using nested loops.

Not required.

Mistake 3

Wrong secondary diagonal formula.

Correct:

`mat[i][n-1-i]`

Edge Cases

Single Element

Input:

`[[5]]`

Output:

5

Even Sized Matrix

Input:

4 x 4

No center overlap.

No subtraction required.

Odd Sized Matrix



Input:

5 x 5

Subtract center once.

12. Detailed Dry Run

Input:

```
[
  [1,2,3],
  [4,5,6],
  [7,8,9]
]
```

i	Primary	Secondary	Sum
0	1	3	4
1	5	5	14
2	9	7	30

Center counted twice:

5

Subtract:

$30 - 5 = 25$

Return:

25

13. Common Use Cases (Real-Life / Interview)

Image Processing

Diagonal pixel operations.

Matrix Analysis

Trace calculations.

Grid-Based Algorithms

Special position traversal.

Interview Purpose

Tests:

Matrix Traversal

Index Manipulation



Pattern Recognition

14. Common Traps (Important!)

Trap 1

Using:

$O(n^2)$

nested loops.

Only diagonals are needed.

Trap 2

Forgetting center overlap.

Trap 3

Mixing row and column indices.

15. Builds To (Related LeetCode Problems)

LC 1572 - Matrix Diagonal Sum

↓

LC 867 - Transpose Matrix

↓

LC 566 - Reshape Matrix

↓

LC 54 - Spiral Matrix

↓

LC 48 - Rotate Image

These strengthen matrix traversal skills.

16. Alternate Approaches + Comparison

Approach	Time	Space	Interview Rating
Nested Loops	$O(n^2)$	$O(1)$	Fair
Direct Diagonal Traversal	$O(n)$	$O(1)$	Best

Approach 1: Nested Loops

Visit every cell.

Check whether:



$row == col$

or

$row + col == n-1$

Works.

Unnecessary.

Approach 2: Direct Traversal (Chosen)

Visit only diagonal elements.

Most efficient.

17. Why This Solution Works (Short Intuition)

Each row contributes exactly:

One primary diagonal element

One secondary diagonal element

Adding both diagonals covers all required cells.

For odd-sized matrices, the center belongs to both diagonals, so subtracting it once prevents double counting.

18. Variations / Follow-Up Questions

Follow-Up 1

Return primary diagonal sum only.

Follow-Up 2

Return secondary diagonal sum only.

Follow-Up 3

Find maximum diagonal sum.

Follow-Up 4

Compute matrix trace.

Related concept:

Primary diagonal only

Interview Takeaway

The key pattern is:



Matrix Index Pattern Recognition

Remember:

Primary Diagonal:

$row == col$

and

Secondary Diagonal:

$row + col == n - 1$

These formulas appear repeatedly in matrix interview problems and are worth memorizing.